Project Proposal

Industrial Metal Detection System for Production Lines

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Abstract

This project proposes an automated metal contaminant detection system for industrial production lines, specifically targeting food and pharmaceutical manufacturing environments. The system uses Hall effect sensors for magnetic field distortion detection to identify ferrous metal contaminants in products. An ESP32 microcontroller manages real-time sensor processing and automated rejection via servo-actuated mechanisms. The system demonstrates Industry 4.0 principles through MQTT-based communication with a Raspberry Pi HMI for remote monitoring, recipe management, and quality statistics. This proof-of-concept addresses critical safety requirements while providing hands-on experience with embedded systems, IoT protocols, and industrial automation.

Contents

1 Problem Statement

In food and pharmaceutical manufacturing, metallic contamination poses severe health risks and regulatory compliance challenges. Metal fragments from processing equipment can contaminate products, leading to:

- Consumer safety hazards (injuries, choking)
- Regulatory violations and production shutdowns
- Costly product recalls

This project develops a proof-of-concept that demonstrates fundamental detection principles using embedded systems and networked communication.

2 Project Objectives & Requirements

2.1 Core Objectives

1. **Detection:** Reliably detect ferrous metal contaminants using Hall effect magnetic field distortion
2. **Automation:** Implement automated rejection mechanism for contaminated items
3. **Communication:** Establish MQTT-based bidirectional communication between ESP32 and Raspberry Pi
4. **Monitoring:** Create functional web-based HMI for remote system monitoring and control
5. **Flexibility:** Implement recipe system for different product sensitivity requirements

3 Technical Methodology

3.1 Metal Detection Principle

Magnetic Field Distortion Method:

1. Neodymium magnet creates uniform magnetic field across conveyor belt
2. Analog Hall effect sensor measures magnetic flux density opposite the magnet
3. When ferrous metal passes through field:
    - Metal distorts magnetic field lines
    - Hall sensor detects voltage deviation from baseline
    - Deviation magnitude indicates metal quantity
4. ESP32 applies threshold-based decision logic

3.2 Operational Sequence

1. **Initialization:** Load recipe, connect MQTT, calibrate baseline, start belt
2. **Detection:** IR sensor detects item → stop belt over Hall sensor
3. **Measurement:** Acquire multiple Hall sensor samples, calculate average
4. **Decision:** Compare against threshold → PASS/FAIL determination
5. **Action:** Publish result via MQTT, resume belt
6. **Rejection:** If FAIL → second IR triggers servo ejection at end of belt
7. **Monitoring:** Update statistics and transmit live data to HMI

3.3 Hardware Components

- **Controller:** ESP32 (sensor interface, MQTT client, motor control)
- **HMI Host:** Raspberry Pi 4 (MQTT broker, Node-RED dashboard, data logging)
- **Sensors:** Analog Hall effect sensor, 2× IR proximity sensors, OLED display
- **Actuators:** DC motor (belt drive), Servo motor (ejection mechanism)
- **Detection:** Neodymium magnet (20mm), play-doh test items with metal screws

4 Communication Protocol

4.1 MQTT Topic Structure

ESP32 → Raspberry Pi (Published):

- `production/status` → System state and active recipe
- `production/detection` → Detection results per item
- `production/statistics` → Cumulative pass/fail counts
- `production/sensor/hall` → Live Hall sensor voltage readings

Raspberry Pi → ESP32 (Subscribed):

- `control/command` → Start/stop/calibrate commands
- `control/recipe/active` → Recipe selection
- `control/recipe/params` → Threshold, speed, sample count parameters

4.2 Recipe Parameters

Each recipe contains tunable parameters:

- **Threshold (V):** Detection sensitivity for metal presence
- **Belt Speed (%):** PWM duty cycle for DC motor
- **Sample Count:** Number of Hall sensor readings to average
- **Auto-Reject:** Enable/disable automatic ejection

5 HMI Design

5.1 Dashboard Features

- **Real-Time Monitoring:** Live Hall sensor voltage, threshold indicator, system status
- **Statistics Panel:** Total items, pass/fail counts, defect rate percentage
- **Recipe Management:** Load/save recipes, adjust parameters via sliders
- **Control Panel:** Start/stop buttons, emergency stop, manual calibration trigger
- **Event Log:** Recent detection results with timestamps

5.2 Implementation

- **Platform:** Node-RED with Dashboard nodes on Raspberry Pi
- **Advantages:** Visual programming, built-in MQTT, rapid development, web-based access

5.3 Conceptual HMI Layout

```
+==================================================================+
| [FACTORY] INDUSTRIAL METAL DETECTION SYSTEM |
+==================================================================+
| |
| STATUS: ⋆ RUNNING Recipe:  [Cake-1 v] |
| [STOP] [CALIBRATE] |
| |
+==================================================================+
| STATISTICS | LIVE READINGS |
+--------------------------+---------------------------------------+
| | |
| Total:  1,247 | Hall Sensor:  |
| Passed:  1,189 | ########-- 2.45V |
| Failed:  58 | |
| Defect:  4.65% | Threshold:  -----+---- |
| | 2.10V |
| [LAST HOUR] | |
| Passed:  89 | Status:  PASS ✓ |
| Failed:  4 | |
| | Belt Speed:  65% |
| | #############------- |
| | |
+==================================================================+
| RECIPE PARAMETERS |
+------------------------------------------------------------------+
| |
| Sensitivity:  [===⋆=====] 2.10V |
| Belt Speed:  [======⋆==] 65% |
| Sample Count:  [===⋆=====] 15 samples |
| Auto-Reject:  [⋆ON] [OFF] |
| |
| [SAVE] [LOAD] [NEW RECIPE] |
| |
+==================================================================+
| RECENT EVENTS |
+------------------------------------------------------------------+
| 14:23:45 - Item #1247 PASSED |
| 14:23:38 - Item #1246 FAILED - Metal detected |
| 14:23:31 - Item #1245 PASSED |
| |
+==================================================================+
```

Figure 1: HMI Dashboard Terminal-Style Mockup

6 Risk Analysis & Mitigation

Risk	Impact	Mitigation Strategy
Hall sensor insufficient sensitivity	High	Early testing with different magnet distances; adjust positioning; use stronger magnet if needed
MQTT connection drops	Medium	Implement auto-reconnection logic; buffer critical data locally; use appropriate QoS levels
Servo timing misalignment	Low	Use second IR sensor for confirmation; add adjustable delay parameter in recipe
Belt slippage or speed inconsistency	Medium	Calibrate belt tension; add reference marks; adjust motor PWM in recipe parameters
Development timeline overrun	Medium	Implement MVP first (basic detection + MQTT); add features incrementally; maintain modular code

Table 1: Risk Analysis and Mitigation Strategies